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A Low Noise 10-Bit 59.41dB SNR SAR ADC Using Chopper Comparator for Biomedical Applications

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Abstract. This paper presents a low noise 10-bit 59.51 dB SNR successive-approximation-register (SAR) analog-digital-inverter (ADC) using the chopper comparator for biomedical applications. The proposed SAR ADC uses a low-power two-stage chopper amplifier (TSCA) to achieve low noise and low power consumption. However, the TSCA is attached to an unlikely offset caused by the process variation, which leads to ripple appearing at the output. Therefore, the proposed chopper comparator uses an offset cancellation loop to mitigate the output ripple. In addition, the TSCA with a self-calibrating input stage is used in the proposed comparator to achieve the lowest delay slope and susceptibility to supply the circuit and have common mode fluctuations compared to prior works. The proposed SAR ADC, which was simulated in a 180 nm CMOS technology, occupies only 0.457 mm² active area. According to simulation results, the prototype SAR ADC achieves a signal-noise ratio (SNR) of 59.41 dB, a signal-to-noise and distortion ratio (SINAD) of 59.06 dB, and an effective number of bit (ENoB) of 9.52 bit while draws a power of 97 μ W leading to the Figure-of-merit (FoM) of 24.1 dB is achieved. Therefore, the proposed SAR ADC is an effective approach for biomedical applications.

Keywords: Chopper comparator · SAR ADC · Offset cancellation loop

1 Introduction

Low power consumption is necessary for portable biomedical sensing applications to run on batteries for extended periods of time. The low-frequency bandwidth of human biosignatures is up to several kHz. Example, an electrocardiogram (ECG) has an amplitude of about 1 mV [1]. An electroencephalogram (EEG) has a frequency range of 0.5 to 150 Hz and an amplitude range of 10 to 100 μ V [2]. The normal amplitude of the local field potential (LFP), which ranges from 1 to 200 Hz, is 1 mV [3]. Since the biosignatures are low-amplitude signals, signal processing must be done first. The analog-to-digital

converter (ADC) is the next important component for signal processing. Thus, the accuracy of the observed biosignatures depends on the performance of the amplifier and the ADC. The successive-approximation-register (SAR) ADC is often used in low-power applications [4].

Comparators are ubiquitous components in mixed-signal systems. The applications such as memories [5], data receivers [6] and ADCs [7] all require high-speed designs with low noise/offset, yet high power consumption and efficiency. The comparator in SAR ADC is very important, since having the need to accurately convert small analog signals into digital information. Therefore, the noise, the offset and speed affect the overall performance of ADC. Dynamic latch type comparators [8] have become very attractive due to fast regeneration time enabled by strong positive feedback and zero static current consumption. Despite having maximum speed at minimum power, small transistor sizes with minimum parasitic loading at the critical nodes are preferred, but this comes with a significantly increased offset [9]. Adding pre-latching gain stages to suppress the offset voltage at the input is a simple approach to address the disadvantages of pre-comparators [10]. However, this leads to increased power consumption since these amplifiers require high gain and low bandwidth. As an alternative, offset compensation techniques have been presented in which digitally controllable capacitors are added to the comparator outputs [11]. Also, several charge pump implementations with additional logic and biasing voltage have been proposed. However, all these approaches degrade the speed of the comparator and its calibration loop, increase the complexity and footprint of the design, and limit robustness.

In this work, a low noise 10-bit 59.51 dB SNR successive-approximation-register (SAR) analog-digital-inverter (ADC) using the chopper comparator is presented for biomedical applications. The proposed SAR ADC uses a low power two-stage chopper amplifier (TSCA) in order to not only improve performance but also achieve low noise as well as low power consumption. However, the TSCA is attached to an unlikely offset caused by the process variation, that leads to ripple appearing at the output. Therefore, the proposed chopper comparator uses an offset cancelation loop to mitigate the output ripple. In addition, the TSCA with self-calibrating input stage is used in the proposed comparator to achieve the lowest delay slope and susceptibility to supply and have common mode fluctuations. The proposed SAR ADC, which was simulated in a 180 nm CMOS technology, occupies only 0.457 mm² active area. According to simulation results, the prototype SAR ADC achieves a signal noise ratio (SNR) of 59.41 dB, a signal to noise and distortion ratio (SINAD) of 59.06 dB, and an effective number of bit (ENoB) 9.52 bit while draws a power consumption of 97 μ W leading to the Figure-of-Merit (FoM) of 24.1 dB is achieved. Therefore, the proposed SAR ADC is an effective approach for biomedical applications.

2 Comparator Architecture

Figure 1 shows the basic schematic structure of the SAR-ADC, which includes the following blocks: Input sampling, comparator, and SAR logic. A sample-and-hold (S/H) circuit samples and holds the input sample. To give the ADC a reliable input, this circuit samples the input voltage at a specific time. The sensed analog voltage and the reference

voltage are compared with a comparator. Depending on whether the input voltage is higher or lower than the reference value, the comparator outputs a digital signal. The SAR block is an iterative digital counter that selects the digital output of the ADC. The SAR consists of flip-flops, where each flip-flop corresponds to one bit in the output of the digital signal. To achieve low power, a two-stage op-amp is used in the comparator. Moreover, the chopper technique is used to modulate the flicker noise or $1/f$ noise to chopping frequency to minimize the noise of the comparator as shown in Fig. 2 at low frequencies. However, due to process variations, the output ripple generated by the chopper output CHout combined with the unlike offset voltage V_{OS} at the input of the comparator degrades the quality of the input signal to the SAR.

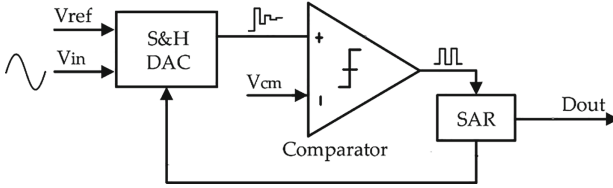


Fig. 1. Basic schematic basic of SAR ADC.

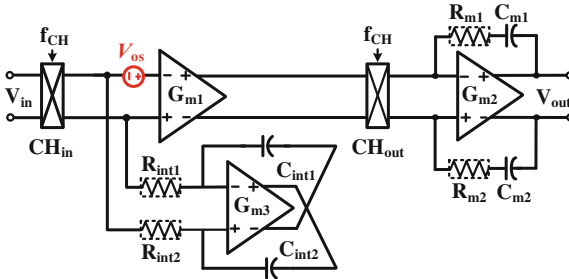


Fig. 2. The schematic of the proposed of comparator in 10-bit SAR ADC.

The proposed chopper comparator using the RC-integrator to mitigate V_{OS} is also shown in Fig. 2. The RC-integrator consists of an op-amp, resistors $R_{int1,2}$, and capacitors $C_{int1,2}$. The V_{OS} at the virtual ground of G_{m1} is filtered out and amplified by the RC-integrator. Due to RC-integrator is implement in cross-couple, then V_{OS} is compensated by the output voltage of RC-integrator. However, employing RC-integrator at the virtual ground, the intrinsic $1/f$ noise of the RC-integrator and the input-stage of TCSA is modulated to the chopping frequency. Therefore, the V_{OS} at G_{m1} is mitigated, that leads the output ripple is also reduced. While the noise source $1/f$ noise of G_{m1} and RC-integrator is addressed by the chopping technique.

The first stage of TCSA, namely G_{m1} , is implemented by the self-bias inverter-based structure, as shown in Fig. 3a. The current-reused is utilized to improve noise efficiency by doubled the transconductance g_m . Moreover, the output (V_{ON1} , V_{OP1}) is fed back for bias voltage, forming the self-bias operation. This way avoids the power consumption from a bias generator block. Due to a fully symmetric differential pair architecture, V_{IN} .

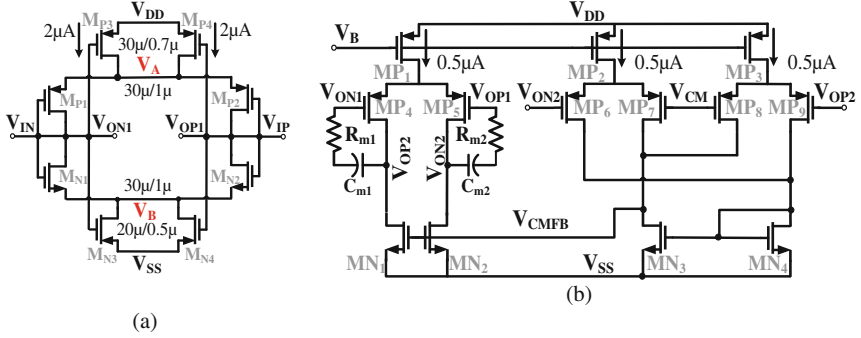


Fig. 3. a) The schematic two stage of the inverter-based amplifier (G_{m1}) with b) common-mode feedback circuit (G_{m2}).

and V_{IP} change by equal opposite amounts from the equilibrium condition, then V_A and V_B are proven not to change. Hence, they are considered as an AC ground node. Moreover, the differential outputs (V_{ON1} , V_{OP1}) generate two voltages with an equal, opposite amount at each node (V_A , V_B) through $M_{P3,4}$ and $M_{N3,4}$. Then, those two voltages are canceled each other. Therefore, V_A and V_B are considered AC ground under the small-signal of both the input (V_{IN} , V_{IP}) and the output (V_{ON1} , V_{OP1}). A common source (CS) amplifier is implemented for the second stage (G_{m2}), as shown in Fig. 3b. The Miller capacitor ($C_{m1,2}$) is inserted to G_{m2} for the stability of the comparator, a common-mode feedback (CMFB) circuitry is used to maintain the output common-mode level at the expected level (V_{CM}). The output of CMFB is fed back to the branches of G_{m2} , which is why common-source stack architecture is used here. The drawback of using the common-source stack is limited output swing due to additional transistors. The schematic of G_{m3} is the two-stage amplifier is shown in Fig. 4. To ensure that the swing of the output of the two-stage amplifier can obtain rail-to-rail, the second stage in thus amplifier uses a class-A amplifier, while the input stage employs a differential amplifier. CMFB set DC voltage level to V_{CM} . The current consumption of two-stage amplifier (including its CMFB) is 11 nA from $V_{DD} = 1.8$ V supply. The MOS transistor's sizing is also presented in Table 1.

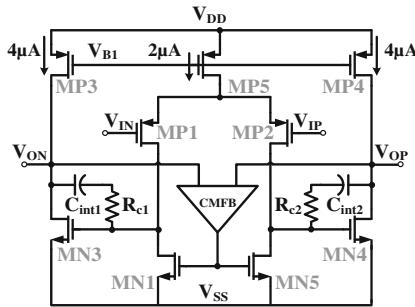


Fig. 4. Schematic of the two-stage op-amp.

Table 1. Parameter W/L of G_{m2} and G_{m3}

G_{m2}		G_{m3}	
Parameter	W/L (μm)	Parameter	W/L (μm)
MN_1, MN_2	0.5/0.5	MN_1, MN_2	8/2
MN_3, MN_4	6/3	MN_3, MN_4	0.5/0.5
MP_1, MP_2, MP_3, MP_4	15/0.7	MP_1, MP_2, MP_5	15/0.7
$MP_5, MP_6, MP_7, MP_8, MP_9$	15/0.7	MP_3, MP_4	60/0.7

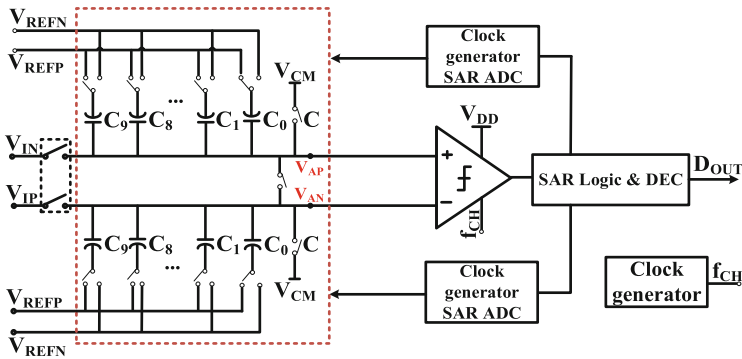
3 Capacitive in SAR ADC 10-Bit

Figure 5 shows a 10-bit differential capacitive DAC. Negative and positive references are used in the differential DAC as defined in Eqs. (1) and (2), respectively.

$$V_{REFN} = V_{CM} - \frac{V_{REF}}{2} \quad (1)$$

$$V_{REFP} = V_{CM} + \frac{V_{REF}}{2} \quad (2)$$

The two input nodes to the comparator are compared in each iteration, so if $V_{AP} - V_{AN} > 0$, then the comparator will output a “1”, otherwise, it produces a “0”. The differential configuration has many advantages over the single-ended configuration, such as better common-mode rejection ratio (CMRR), the reduction of harmonic distortion, and doubling of the input voltage range, but at the cost of a larger layout area, since most components are doubled for the differential configuration. The capacitive DAC uses charge redistribution to convert the digital code coming from the logic SAR into the corresponding analogue voltage level, which is compared by the comparator of the SAR ADC.

**Fig. 5.** The schematic of SAR ADC 10-bit.

4 Simulation Results

In the 180 nm CMOS technology, the fabricated prototype occupies an active chip area of 0.457 mm^2 with $97 \mu\text{W}$ power consumption as shown in Fig. 6. The marked contents of the die micrograph correspond to each sub block of the proposed SAR ADC. At the frequency 500 Hz, the noise voltage when applied to the input of without chopping circuit is elder than with chopping circuit. To verify the function of chopper comparator, an offset voltage of 10 mV is applied for both G_{m1} and G_{m3} then perform Periodic-Steady-State (PSS) simulation tool with an input voltage of zero. Figure 7 shows the output ripple, which is a triangle wave with a frequency of 10 kHz (chopping frequency).

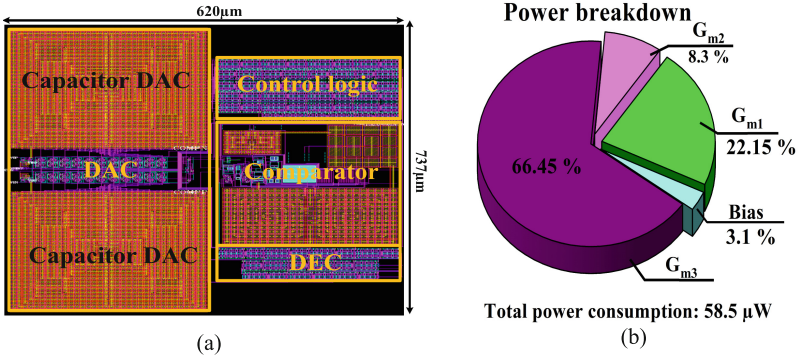


Fig. 6. a) Layout of the SAR ADC 10-bit, b) the power breakdown of the comparator.

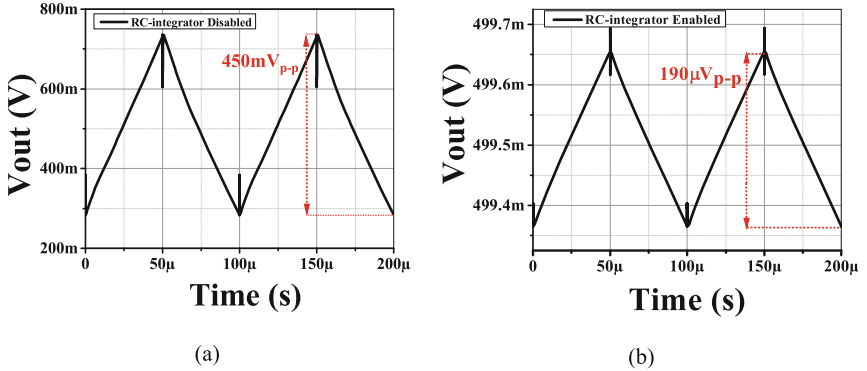


Fig. 7. Simulated output ripple a) without RC-integrator, b) with RC-integrator.

When RC-integrator is disabled and enabled in sequence, the amplitude of the ripple is 440 mV_{pp} and $188 \mu\text{V}_{pp}$. So, the output ripple of TSCA of using RC-integrator circuit in comparator reduces approximately 200(x) compared to without RC-integrator that limits noise and offset in comparator. Measured input-referred noise of the proposed comparator in SAR ADC between without and with chopping is shown in Fig. 8a. The

FFT spectrum shows in Fig. 8b that it achieves an ENoB of 9.52-bit at 1 kHz input frequency and an input single with a peak-to-peak voltage range of 87.86 mV. Table 2 summaries the performance of the proposed architecture and its comparison with other prior art architectures. It is evident that the proposed architecture has a competitive performance in terms of energy efficiency and linearity. The proposed structure achieves a FoM of 24.1 dB at fJ/con.step. With the chopper structure in SAR ADC, [12] uses a supply voltage of 1.2 V, but ENoB achieves only 14.4 bits, while the number of bits is 18 bits, then SAR ADC operates with negligible accuracy. Besides, the power of [13] is only 14.22 μ W low, but the chip area is 12 times the chip area of this work and the power consumption is 97 μ W.

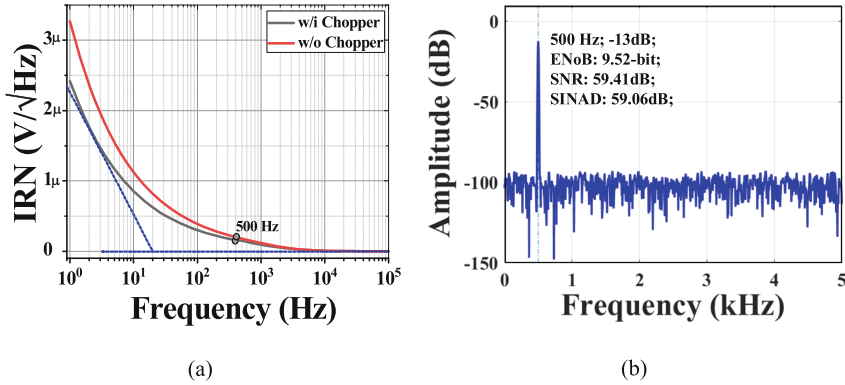


Fig. 8. a) Measured input-referred noise of the proposed comparator in SAR ADC, b) FFT-response of the SAR ADC.

Table 2. Performance comparison.

Parameter	Hirai [12]	Choi [13]	This work
Architecture	Chopper-stabilized LNA	CFIA (with RFC) + Chopping	TSCA + Chopping
Tech. (nm)	130	180	180
Supply (V)	1.2	1.8	1.8
Area (mm ²)	4	12.5	0.457
Power (μ W)	5.5m	14.22	97
IRN (nV/ $\sqrt$ Hz)	2.52 μ	86.6	27.72
ENoB (bit)	14.4	—	9.52
FoM	—	—	24.1

5 Conclusion

A high-speed, low noise TSCA comparator with a dual-sided self-calibrating loop has been presented. The proposed chopper comparator calibration tremendously reduces the input offset approximately 200(x), limited only by the comparator noise, without compromising its speed for significantly increased power. Combined with the implemented multi-stage comparator to enable better optimization between various trade-offs, the highest reported frequency of 10 MHz is realized with only 27.72 μV input noise and offset from a 1.8 V supply. The prototype occupies a total area of only 0.457 mm². In summary, the proposed circuit is an ideal candidate for any high speed, low noise/offset, power-area-efficient mixed-signal system and can be adapted to any comparator structure.

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